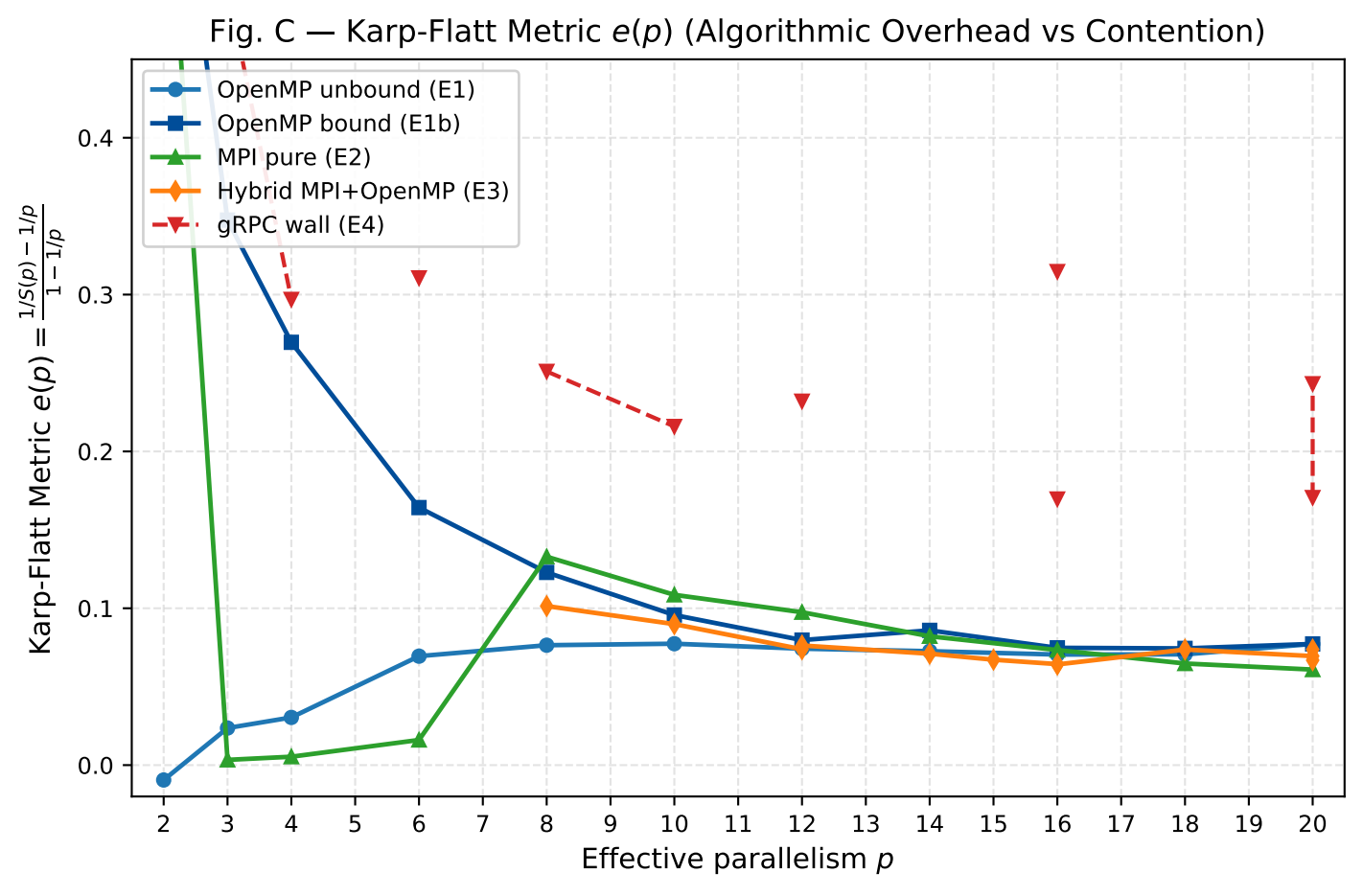
Fig. C — Karp-Flatt Metric $e(p)$ (Algorithmic Overhead vs Contention)
OpenMP unbound (E1)
OpenMP bound (E1b)
MPI pure (E2)
Hybrid MPI+OpenMP (E3)
gRPC wall (E4)
Karp-Flatt Metric $e(p) = \frac{1/S(p) - 1/p}{1 - 1/p}$
Effective parallelism p
0.0
0.1
0.2
0.3
0.4
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20